

RT-03 · Sun 16 Aug 2026

4 chapters across 3 subjects

Weightage from your MSBSHSE 11th marks distribution

"Top-6" = highest-ROI chapters per subject

PHYSICS

1 chapter · 6 marks (8 with option) · 8.6% of Physics

Motion in a Plane

6 / 8 marks

8.6% of Physics

#3 priority chapter in Physics

- Scalars vs. vectors — representation, types (equal, negative, unit, null, position, displacement, co-initial, collinear)
- Vector addition — triangle law, parallelogram law, polygon law
- Subtraction of vectors
- Resolution of a vector into rectangular (perpendicular) components
- Multiplication of vectors — scalar (dot) product and vector (cross) product, geometrical meaning
- Relative velocity — velocity of one body with respect to another, in a plane
- Projectile motion — horizontal projectile and oblique/angular projectile: time of flight, maximum height, horizontal range, velocity at any instant
- Uniform Circular Motion (UCM) — angular displacement, angular velocity, angular acceleration, centripetal acceleration, period & frequency, $v = r\omega$

Why it matters: this is only physics chapter on RT-03, but it's the 3rd-highest weightage chapter in the entire Physics syllabus — behind only Laws of Motion and Optics. This is the whole test for Physics, so it deserves full depth, not a skim.

CHEMISTRY

2 chapters · 7 marks (10 with option) · 10% of Chemistry

Basic Analytical Techniques

2 / 3 marks

2.9% of Chemistry

⚠️ Lowest-weightage chapter in Chemistry

- Introduction and scope of analytical techniques
- Chromatography — principle (differential migration), types: paper chromatography, thin layer chromatography (TLC), column chromatography
- Rf value and its significance
- Basic instrumental concepts — colorimetry / spectrophotometry (qualitative introduction)
- pH-metry and potentiometry — brief introduction

Time budget: this is the single lightest chapter in all of Chemistry. Read it once for concepts and definitions — don't over-invest drilling numericals here, there mostly aren't any.

Structure of Atom (RT-03 scope only)

5 / 7 marks*

7.1% of Chemistry

#6 priority chapter in Chemistry

- Introduction — discovery of the electron, proton, and neutron (subatomic particles)
- Thomson's atomic model (plum-pudding model) — postulates and limitations
- Rutherford's atomic model (alpha-particle scattering experiment) — postulates and limitations
- Electromagnetic spectrum — wave characteristics: wavelength, frequency, wave number, speed of light relation
- Bohr's atomic model — postulates, energy levels, limitations

⚠️ Scope note: the test plan lists only the portion above for RT-03. The full "Structure of Atom" chapter continues into quantum numbers, orbitals, electronic configuration (Aufbau, Pauli, Hund's rule) — that's **not** RT-03's scope, so the 5/7-mark weightage (*) reflects the whole chapter, not just this slice. Confirm the cutoff with your tuition teacher before assuming the rest is safe to skip for 16 Aug.

Determinants and Matrices

7 / 10 marks

8.75% of Maths

 #1 priority chapter in all of Maths

- Determinants — definition, order-2 and order-3 determinants, expansion
- Properties of determinants
- Minors and cofactors
- Applications of determinants — area of a triangle, collinearity of three points, Cramer's rule, consistency/inconsistency of a system of equations
- Matrices — definition, order, types (row, column, square, diagonal, scalar, identity/unit, null/zero, symmetric, skew-symmetric)
- Algebra of matrices — equality, addition, subtraction, scalar multiplication, multiplication of matrices
- Transpose of a matrix and its properties
- Adjoint of a matrix
- Inverse of a matrix (via the adjoint method)
- Elementary row/column transformations
- Application — solving a system of linear equations by the matrix method

Why it matters: this is the single highest-weightage chapter in the entire 18-chapter Maths syllabus — ahead of Conic Sections, Trig-II, and Permutations & Combinations. Miss this and you're leaving the single biggest Maths chunk on the table.

MISMATHE'S READ ON RT-03

- **Two chapters carry the test:** Motion in a Plane (Physics) and Determinants & Matrices (Maths) are each top-3-in-subject weightage chapters. These deserve your deepest practice time — full depth, timed numericals, PYQ-style questions.
- **Structure of Atom is high-value but partial-scope** — don't let it balloon into the full chapter. Nail exactly the 5 topics listed above for 16 Aug, then circle back for orbitals/configuration later.
- **Basic Analytical Techniques is a quick box to tick** — lowest weightage chapter in Chemistry, mostly definitions. One focused read is enough; don't burn drilling hours here that Determinants & Matrices or Motion in a Plane could use instead.
- **Suggested order:** Determinants & Matrices → Motion in a Plane → Structure of Atom → Basic Analytical Techniques (heaviest and most calculation-heavy first, while your recall energy is freshest).